Jinil Kim

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RESEARCH INTERESTS

- *Computational Neuroscience* (principles linking brain and behavior; inferring their shared latent states)
- *NeuroAI and AI for Neuroscience* (brain foundation models; neuro-inspired AI)
- *Computational Psychiatry* (model-derived markers; predicting intervention response)

EDUCATION

Seoul National University — Seoul, South Korea
B.S. Computer Science · B.A. Psychology · B.A. Linguistics — Mar 2021 – Exp. Feb 2027
GPA: 4.18 / 4.3 (3.97 / 4.0) · 9 semesters (Mandatory military service, Jul 2023 – Jan 2025)

The University of Hong Kong — Hong Kong SAR
Summer Research Program, Department of Psychology — Jun 2025 – Aug 2025

PUBLICATIONS & MANUSCRIPTS

Workshop Papers

- **Kim, J.**, & Ryu, J. (2026). *Conversational relevance in the conjunction fallacy is represented independently in language models.* Accepted for Conference on Neural Information Processing Systems (NeurIPS) 2026 Workshop on Linguistic Principles for Foundation Models, Sydney, Australia. [paper]
- **Kim, J.**, Cho, A. M., Seo, J., & Cha, J. (2026). *Inferring individualized color-vision distortions from fMRI hue-representation geometry.* Presented at International Conference on Machine Learning (ICML) 2026 Workshop on Structured Data for Health, Seoul, Korea. [paper]

Under Review

- **Kim, J.**, Jeong, J., Yun, H., Oh, M.-H., & Ahn, W.-Y. (2026). *Inverse reinforcement learning with state-dependent foresight for human decision making.* Under review at the International Conference on Learning Representations (ICLR) 2027.

In Preparation

- **Kim, J.**, Cho, A. M., Seo, J., & Cha, J. *Preserved categorical identification with impaired hue interpolation in color vision deficiency motivates a cortically derived correction filter.*
- **Kim, J.**, & Yao, Y.-W. *Depression is associated with weaker evidence accumulation and greater caution during social decision-making.*

RESEARCH EXPERIENCES

Connectome Lab — Seoul National University
Undergraduate RA (Advisor: Prof. Jiook Cha) — Apr 2025 – Present

- Led a project characterizing distortions of cortical hue geometry in color vision deficiency from fMRI. Translated them into personalized filters that moved hue-discrimination thresholds to the control range.
- Leading uncertainty evaluation of generative EEG foundation models, showing that distributional scores catch distortions that point metrics miss and that seed variance can flag unreliable channels.
- Training brain foundation models with literature-derived priors via text alignment or by using each region's functional profile as its positional encoding.

Computational Clinical Science Lab — Seoul National University
Undergraduate RA (Advisor: Prof. Woo-Young Ahn) — Apr 2025 – Present

- Led the development of SC-AIRL, which adds state-dependent foresight to AIRL and raises accuracy on held-out human actions by 9–10 points. Inferred planning depth correlated with anxiety and impulsivity.
- Testing whether fMRI, EEG, HRV, and survey data jointly predict alcohol-use severity ($n = 60$) and yield interpretable markers, using multiblock sparse PLS.

MIND Lab — University of Hong Kong
Visiting Researcher (Advisor: Prof. Yuan-Wei Yao) — Jun 2025 – Aug 2025

- Fit stationary and time-varying drift-diffusion models to social decisions ($N = 351$), linking depression to weaker evidence accumulation ($d = 0.30$), greater caution ($d = 0.40$), and later reward-emotion integration.

CONFERENCE PRESENTATIONS

- **Kim, J.**, Cho, M., Seo, J., & Cha, J. (2026). *fMRI decoding reveals intact neural color representations in color vision deficiency.* Organization for Human Brain Mapping (OHBM) 2026 Annual Meeting, Bordeaux, France. [Poster]
- **Kim, J.**, & Cha, J. (2026). *A framework decomposing individual arousal predictions into visual-concept and neural contributions.* Submitted to Computational and Systems Neuroscience (Cosyne) 2027.

SCHOLARSHIPS & HONORS

KOSAF Humanity 100 Years Undergraduate Scholarship ($12,500) Mar 2025 – Jul 2026

- National merit scholarship from the Korean Ministry of Education; full tuition and living stipend.

Best Poster Award, SNU Undergraduate Research Program Jan 2026
SNU Undergraduate Research Grant ($3,500) Apr 2025
SNU Student-Proposed Liberal Arts Course Contest (1st Place) Jul 2023

- Proposed course: *Interdisciplinary Explorations with a Chatbot.*

SNU Academic Excellence Scholarship Mar 2022 – Jun 2023

RELEVANT COURSEWORK

Neuroscience	Neuroscience · Data Science in Human Neuroimaging · Clinical Neuropsychology · Cognitive Processes · Computational Modeling in Experimental Psychology · Brain, Mind, and Behavior
Computer Science	Data Structures · Algorithms · Computer Architecture · Logic Design · Computer Programming · Computational Linguistics
Mathematics & Statistics	Linear Algebra · Discrete Mathematics · Calculus I–II · Psychological Statistics · Advanced Psychological Statistics

ACADEMIC SERVICES

Conference Reviewer: NeurIPS 2026 Workshop on Foundation Models for the Brain and Body
Volunteer: ICML 2026

TEACHING EXPERIENCE

Basic Computing: First Adventure in Computing Seoul National University
Teaching Assistant (Instructors: Yamada Akihiko, Heajung Min) Autumn 2025 · Spring 2026
Taught Python lab sessions for local and international students, including an English-medium section in Spring 2026.

LANGUAGES

Programming: Python (advanced), R, C++, Java (proficient), Stan (intermediate)